

Brianna Quinn & [redacted] [redacted]

DS4300 – Homework 5

Neo4j Queries:

Create Node:

```
LOAD CSV WITH HEADERS FROM 'file:///songs.csv' AS row
CREATE (:Song {
  track_id: row.track_id,
  artists: row.artists,
  album_name: row.album_name,
  track_name: row.track_name,
  popularity: toInteger(row.popularity),
  duration_ms: toInteger(row.duration_ms),
  explicit: row.explicit,
  danceability: toFloat(row.danceability),
  energy: toFloat(row.energy),
  key: toInteger(row.key),
  loudness: toFloat(row.loudness),
  mode: toInteger(row.mode),
  speechiness: toFloat(row.speechiness),
  acousticness: toFloat(row.acousticness),
  instrumentalness: toFloat(row.instrumentalness),
  liveness: toFloat(row.liveness),
  valence: toFloat(row.valence),
  tempo: toFloat(row.tempo),
  time_signature: toInteger(row.time_signature),
  track_genre: row.track_genre
})
```

Create Edge

```
LOAD CSV WITH HEADERS FROM 'file:///edges.csv' AS row
MATCH (a:Song {track_id: row.source})
MATCH (b:Song {track_id: row.target})
CREATE (a)-[:SIMILAR_TO {
  distance: toFloat(row.distance)
}]->(b)
```

Recommendation Query

```
MATCH (seed:Song)-[r:SIMILAR_TO]->(rec:Song)
WHERE (seed.artists CONTAINS 'the strokes'
OR seed.artists CONTAINS 'regina spektor')
AND NOT rec.artists CONTAINS 'the strokes'
AND NOT rec.artists CONTAINS 'regina spektor'
AND (
  rec.track_genre CONTAINS 'rock'
OR rec.track_genre CONTAINS 'indie'
OR rec.track_genre CONTAINS 'alternative'
OR rec.track_genre CONTAINS 'punk'
OR rec.track_genre CONTAINS 'garage'
)
RETURN
rec.track_name AS recommended_song,
rec.artists AS artist,
rec.album_name AS album,
rec.track_genre AS genre,
count(*) AS supporting_links,
avg(r.distance) AS avg_distance
ORDER BY avg_distance ASC, supporting_links DESC
LIMIT 15
```